Tata Power EV Charging Infrastructure Analysis

Presented by : Avijit Kumar

Business Performance Analysis & Expansion Strategy

Project Objectives

01 *Analyze Business Performance*

Evaluate charging demand, revenue generation, energy consumption, and overall network performance to identify growth trends

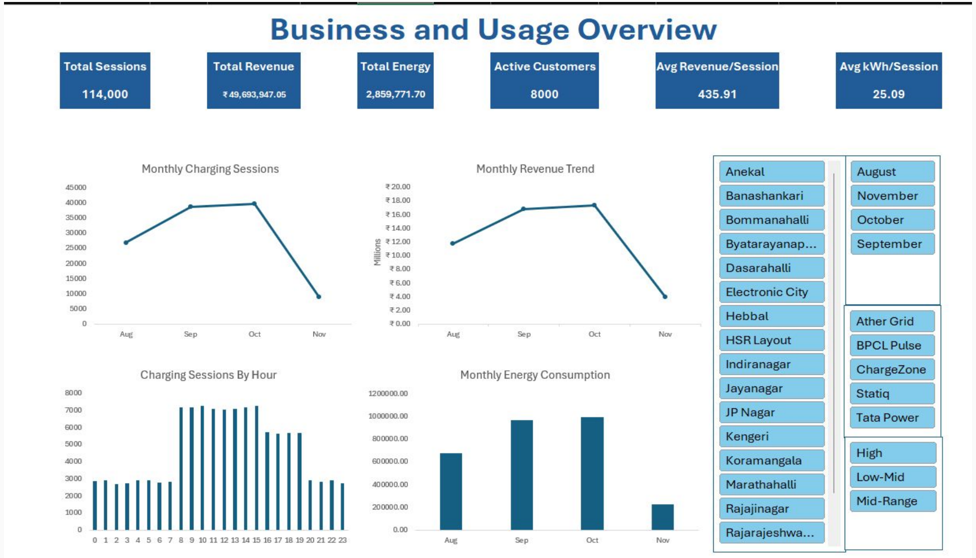
02 *Measure Operational Efficiency*

Assess station utilization, plug efficiency, customer charging behavior, and infrastructure performance across districts.

03 *Support Strategic Expansion*

Analyze ROI, payback period, and financial metrics to identify high-potential locations for future EV charging expansion

Business & Usage Overview



Dashboard Objective

Monitor charging demand, revenue generation, energy consumption, and customer activity to evaluate the overall performance of the EV charging network.

Key KPIs

- **Total Charging Sessions-** 114,000
- **Total Revenue-** ₹49.69 Million
- **Total Energy Delivered-** 2.86 Million kWh
- **Active Customers-** 8,000
- **Avg. Revenue per Session-** ₹435.91
- **Avg. Energy per Session-** 25.09 kWh

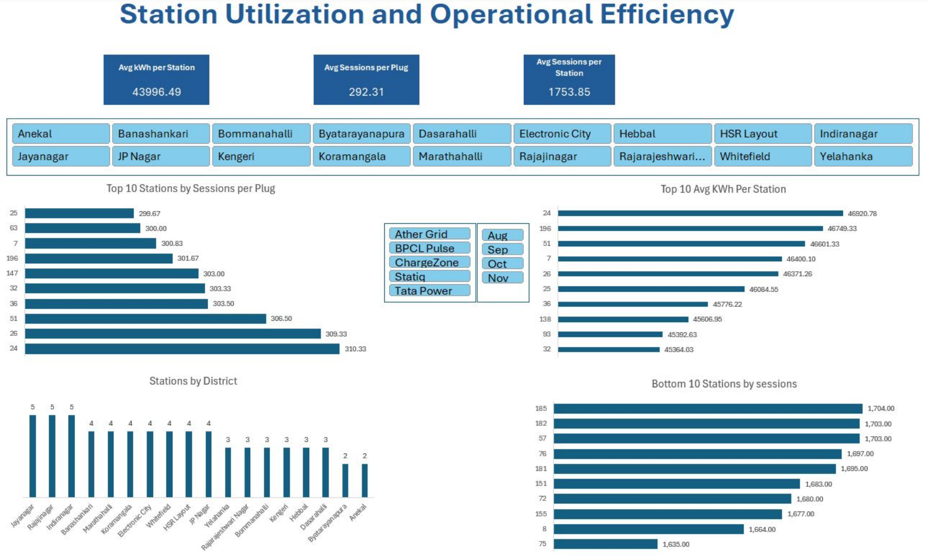
Key Insights

- 114,000 charging sessions indicate strong network adoption.
- Revenue reached ₹49.69M, reflecting healthy business performance.
- October recorded the highest charging activity and revenue.
- Average energy per session remained consistent at 25.09 kWh.
- 8,000 active customers demonstrate strong user engagement

Conclusion

The EV charging network demonstrates strong demand, steady revenue growth, and consistent customer engagement, indicating a scalable and commercially viable infrastructure.

Station Utilization & Operational Efficiency



Dashboard Objective

Evaluate charging station utilization, plug efficiency, and district-wise infrastructure performance to identify operational bottlenecks and optimize resource allocation.

Key KPIs

- **Avg. Sessions per Station -** 1,754
- **Avg. Sessions per Plug -** 292
- **Avg. Energy per Station -** 43,996.49 kWh

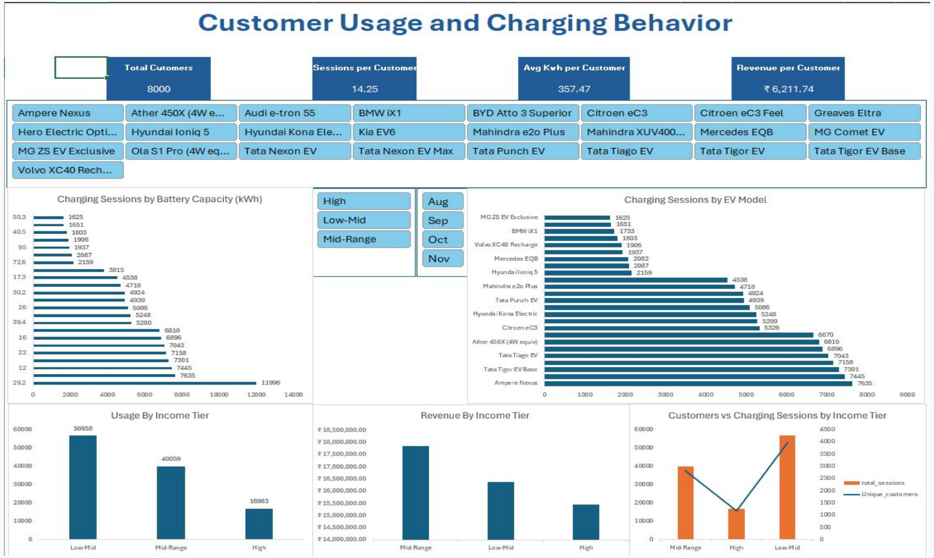
Key Insights

- Each station handled an average of 1,754 charging sessions.
- Charging plugs were efficiently utilized with 292 sessions per plug.
- A few stations consistently outperformed others in usage and energy delivered.
- Station demand varied across districts, highlighting opportunities for infrastructure optimization.
- High-performing stations provide a benchmark for future expansion planning.

Conclusion

The charging network operates efficiently, with strong station utilization and plug performance. Optimizing low-performing locations and expanding capacity in high-demand areas can further improve operational efficiency.

Customer Usage & Charging Behavior



Dashboard Objective

Analyze customer charging behavior across income groups, EV models, and battery capacities to identify high-value customer segments and usage patterns.

Key KPIs

- **Unique Customers:** 8,000
- **Avg. Sessions per Customer:** 14.25
- **Avg. Energy per Customer:** 357.47 kWh
- **Avg. Revenue per Customer:** ₹6,211.74

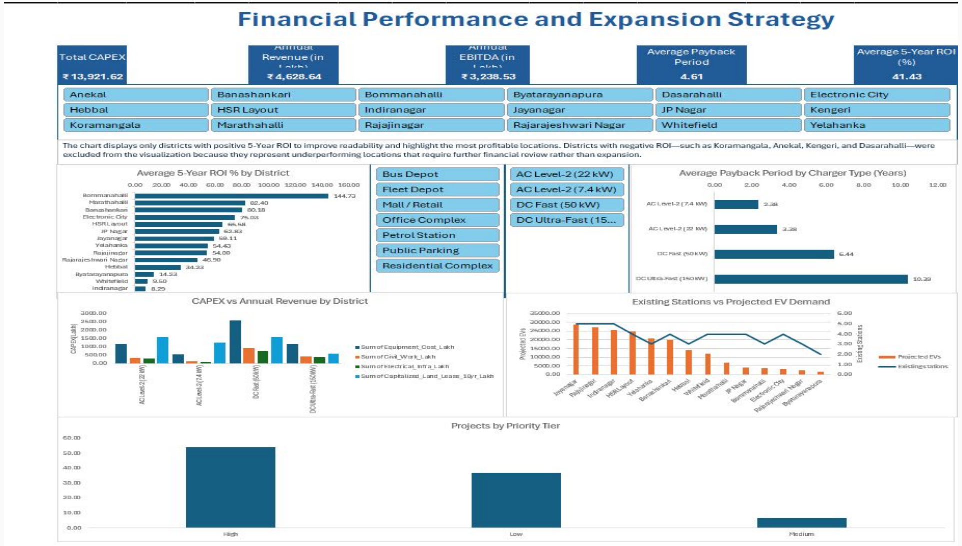
Key Insights

- Customers averaged over 14 charging sessions, indicating strong repeat usage.
- Low-Mid income customers accounted for the highest charging activity.
- Charging demand varied across EV models and battery capacities.
- Revenue per customer remained consistently high, reflecting strong customer value.
- Customer segmentation reveals opportunities for targeted marketing and loyalty programs.

Conclusion

Customer analytics indicate strong engagement and repeat charging behavior. Understanding usage patterns across customer segments can help improve retention, personalize services, and drive future demand.

Financial Performance & Expansion Strategy



Dashboard Objective

Evaluate the financial viability of EV charging projects using investment, profitability, ROI, and payback metrics to support data-driven expansion decisions.

Key KPIs

- **Total CAPEX:** ₹13,921.62 Lakh
- **Annual Revenue:** ₹4,628.64 Lakh
- **Annual EBITDA:** ₹3,238.53 Lakh
- **Avg. Payback Period:** 4.61 Years
- **Avg. 5-Year ROI:** 41.43%

Key Insights

- The network generated ₹4,628.64 lakh in annual revenue against a CAPEX of ₹13,921.62 lakh.
- Projects achieved a healthy average 5-year ROI of 41.43%.
- The average payback period of 4.61 years indicates strong investment potential.
- ROI and profitability vary across districts, highlighting priority areas for expansion.
- Financial metrics support continued investment in high-performing locations.

Conclusion

The financial analysis confirms that the EV charging infrastructure is commercially viable. Prioritizing districts with higher ROI and shorter payback periods will maximize returns and support sustainable network expansion.

Overall Insights

Business Performance

- 114,000 charging sessions
- ₹49.69M revenue
- Strong month-on-month growth

Operational Efficiency

- High station utilization
- Efficient plug usage
- Demand concentrated at top-performing stations

Customer Insights

- 8,000 active customers
- Strong repeat charging behavior
- Low-Mid income segment drives the highest usage

Financial Performance

- 41.43% average ROI
- 4.61-year payback period
- Positive EBITDA supports future expansion

Strategic Recommendations

Operations

- Increase capacity at high-demand stations.
- Improve utilization of low-performing stations.

Customer

- Launch loyalty programs.
- Encourage off-peak charging through targeted incentives

Expansion

- Prioritize districts with higher ROI and shorter payback periods.
- Expand infrastructure based on projected EV demand.

Financial

- Continuously monitor ROI and operating costs.
- Allocate capital to high-performing projects.

Conclusion

Final Takeaways

- The EV charging network demonstrates strong operational and financial performance.
- Customer adoption and charging demand continue to grow steadily.
- Efficient station utilization and healthy ROI support continued investment.
- Data-driven expansion will help maximize profitability and improve network coverage